

Supporting Information

A Reproducible Event-Fixed Protocol for Evaluating Financial Stress Detectors

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This file collects the sensitivity and ablation material referenced from the main text: event-window sensitivity (Table S1, Figure S1), the composite weight ablation (Table S2), the sign-convention ablation for the persistence statistic (Table S3), and the confirmation-window sensitivity (Figure S2). All values are produced by the reproducibility package accompanying the article.

Table S1. Event-window sensitivity for pointwise precision, recall, F1, and FPR.

Window	Method	Prec.	Recall	F1	FPR
± 20	Vol.	0.390	0.396	0.393	0.061
± 20	Hybrid	0.545	0.216	0.309	0.018
± 30	Vol.	0.499	0.344	0.407	0.052
± 30	Hybrid	0.692	0.186	0.293	0.012
± 45	Vol.	0.571	0.266	0.363	0.048
± 45	Hybrid	0.815	0.148	0.251	0.008
± 60	Vol.	0.640	0.225	0.333	0.044
± 60	Hybrid	0.886	0.122	0.214	0.005

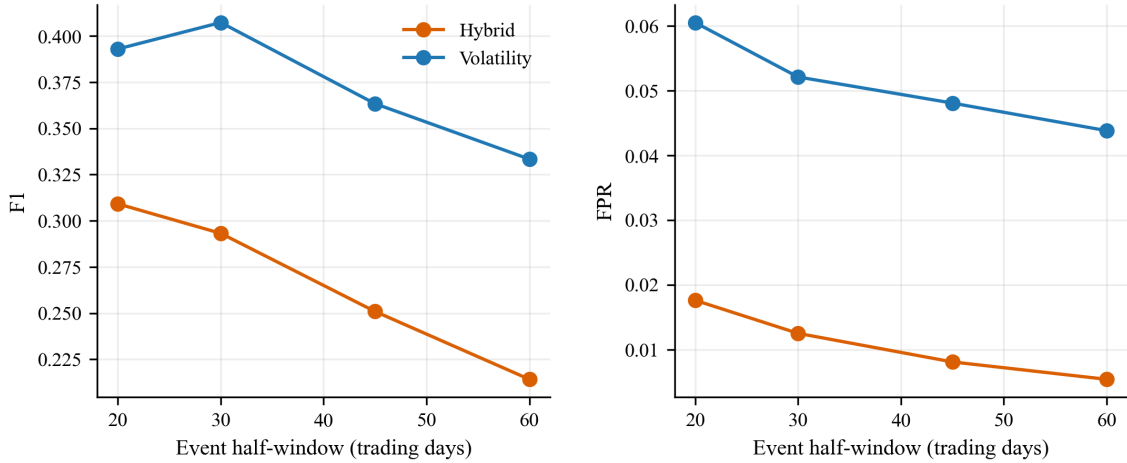


Figure S1. Event-window sensitivity across ± 20 , ± 30 , ± 45 , and ± 60 trading-day windows.

Table S2. Composite weight ablation. Each row recomputes x_t , the MMD and κ components, and the hybrid. Events are counted out of 13.

(w_r, w_σ, w_z)	Prec.	Recall	F1	FPR	Delay	Events
(0.5, 0.3, 0.2) default	0.886	0.122	0.214	0.005	6.8	8
(1/3, 1/3, 1/3) equal	0.886	0.122	0.214	0.005	6.8	8
(0.2, 0.3, 0.5) reversed	0.886	0.122	0.214	0.005	6.8	8
(0.6, 0.2, 0.2)	0.882	0.122	0.214	0.006	6.8	8
(0.4, 0.4, 0.2)	0.886	0.122	0.214	0.005	6.8	8
(0.5, 0.2, 0.3)	0.886	0.122	0.214	0.005	6.8	8
(0.2, 0.6, 0.2)	0.900	0.135	0.235	0.005	4.4	8
(1, 0, 0) returns only	0.845	0.131	0.227	0.008	0.8	8
(0, 1, 0) volatility only	0.815	0.121	0.210	0.009	1.4	7
(0, 0, 1) z-score only	0.886	0.122	0.214	0.005	6.8	8

Table S3. Confirmation-term ablation for the persistence statistic. The low- κ variant is the direction predicted by critical-slowing-down theory.

Confirmation term	Prec.	Recall	F1	FPR	Delay	Events
$ z_\kappa $ two-sided (default)	0.886	0.122	0.214	0.005	6.8	8
$\max(-z_\kappa, 0)$ low κ	0.875	0.114	0.202	0.006	6.7	7
$\max(+z_\kappa, 0)$ high κ	0.874	0.141	0.242	0.007	1.6	7

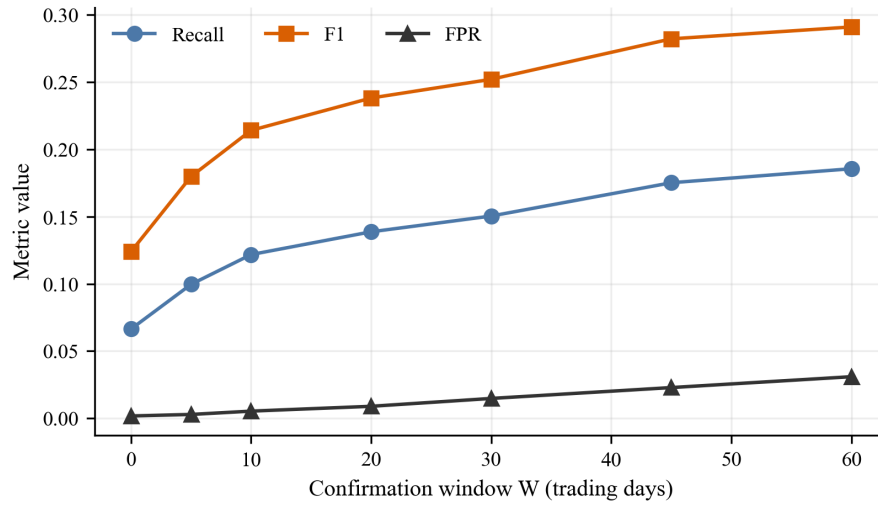


Figure S2. Hybrid confirmation-window sensitivity showing how the retrospective confirmation window changes F1 and FPR.